

Lab 3: Probability and Distributions

Instructor notes – Part 1: prep reading, Part 2: podium/circulating cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – an answer key and the reasoning behind each Act. **Part 2** (after the page break, large print) is the copy to have open while circulating during lab.

Schedule note: Labor Day cancels this week’s Monday meeting, so this lab runs with no prior lecture on Bayes’ rule – Wednesday’s lecture comes after, and the bestiary of distributions has moved to next Monday (Week 4). **Design note:** Act 1 is deliberately discovery-based rather than “apply the formula you were just taught” – students derive the PPV calculation themselves from plain counting before ever seeing it written as $P(H | D)$. **No laptops this lab** – all three Acts are pen-and-paper (a calculator is enough); Acts 2 and 3 use pre-rendered figures and a printed z-table instead of `dbinom()`/`dpois()`/`pnorm()`/`qnorm()`, since sampling/evaluating a named distribution isn’t, by itself, enough reason to open a laptop – that’s reserved for weeks with real algorithm-development content.

Suggested opening (~5 min, before releasing Act 1)

Keep this short – a full treatment of conditional probability and Bayes’ rule is Wednesday’s job, not today’s. Two rules are enough to unlock Act 1:

- **Addition:** probabilities of mutually exclusive (disjoint) outcomes add up (e.g., “flagged” splits into “flagged AND really the species” plus “flagged AND not the species” – no overlap, so these two pieces sum to the total flagged).
- **Multiplication / applying a rate:** a probability or rate applied to a count of events gives you the expected count in that subgroup (e.g., if 10 events are truly the species and the classifier catches 98% of them, expect about $10 \times 0.98 \approx 10$ flagged).

That’s it – don’t derive Bayes’ rule or use conditional-probability notation ($P(A | B)$) in this opening; Act 1 gets there on its own, and Wednesday’s lecture gives it a name.

Act 1: camera-trap classifier PPV (discovery-based)

Part 1 (counting). Same numbers as before ($P(\text{species}) = 0.0001$, sensitivity = 0.98, false positive rate = 0.01), $N = 100,000$.

```

N <- 1e5; p_S <- 1e-4; sens <- 0.98; fpr <- 0.01
species      <- N * p_S
not_species  <- N * (1 - p_S)
flag_S       <- species * sens
flag_notS    <- not_species * fpr
data.frame(row = c("Actually the species", "Not the species"),
           flagged = round(c(flag_S, flag_notS)),
           not_flagged = round(c(species - flag_S, not_species - flag_notS)),
           total = c(species, not_species))

```

```

##               row flagged not_flagged total
## 1 Actually the species      10          0     10
## 2      Not the species    1000      98990 99990

```

Answer key (1a-1b): Flagged/true-species ≈ 10 ; flagged/not-species $\approx 1,000$; total flagged $\approx 1,010$. PPV (1b) = $10/1010 \approx 0.0099 \approx 1\%$ – pure counting, no formula.

Answer key (1c): open-ended, but the number should read as “surprisingly low given how good the classifier sounded” – a single flag isn’t enough to trust; the field team should look for corroborating evidence (a second detection, a track, scat) before acting on one image.

Part 2 (formalize). Guided derivation:

```

# numerator: N * p_S * sens ; denominator: N*p_S*sens + N*(1-p_S)*fpr
# N cancels, leaving:
ppv_formula <- (p_S * sens) / (p_S * sens + (1 - p_S) * fpr)
ppv_formula

```

```
## [1] 0.009705853
```

Answer key (2a-2c): numerator = $N \times \text{prevalence} \times \text{sensitivity}$; denominator = $N \times \text{prevalence} \times \text{sensitivity} + N \times (1 - \text{prevalence}) \times \text{false positive rate}$. N cancels, leaving

$$\text{PPV} = \frac{\text{prevalence} \times \text{sensitivity}}{\text{prevalence} \times \text{sensitivity} + (1 - \text{prevalence}) \times \text{false positive rate}} \approx 0.0097$$

This is Bayes’ rule / PPV, arrived at without ever naming it – don’t name it for them either; that reveal is Wednesday’s job. If a group gets stuck on 2a, the hint is: “what did you multiply by 100,000 to get each cell of your table in Part 1? Do that again, but write N instead of 100,000.”

Part 3 (transfer – Site B). Prevalence = 0.01, same sensitivity and false positive rate.

```

p_S_B <- 0.01
ppv_B <- (p_S_B * sens) / (p_S_B * sens + (1 - p_S_B) * fpr)
ppv_B

```

```
## [1] 0.4974619
```

Answer key (3a): $PPV \approx 0.497 \approx 50\%$ – a dramatic jump from Site A’s $\approx 1\%$, same classifier. **(3b):** the intended takeaway is that prevalence dominates PPV every bit as much as the classifier’s own accuracy – a “good” classifier is only as useful as the base rate of what it’s looking for allows.

Watch for: groups who compute the fraction backwards ($P(S)$ instead of $P(F)$ in the denominator of 1b). A $PPV > 0.5$ at Site A is almost always the tell. In Part 2, watch for groups who try to cancel something other than N , or who get stuck because they’re trying to recall “the formula” instead of generalizing their own Part 1 numbers.

Act 2: binomial detections, and when Poisson is a good stand-in

Answer key (1a). The binomial’s full assumption list is exactly four items – fixed N , binary (Bernoulli) outcome per trial, independence across trials, constant p across trials – and a complete answer names all four. What separates a thin answer from a strong one is the “plausible or not” half of the question:

- **Fixed $N = 8$ and binary outcome per night** are true *by construction* here (the deployment length is set in advance; “at least one detection” vs. none is how the variable is defined) – there’s not much to interrogate in either one.
- **Independence** and **constant p** are the two actually worth scrutinizing, and a strong answer says so. Independence could break down if a malfunction, a multi-night weather event, or the animal revisiting a scent-marked area links one night’s outcome to the next’s. Constant p could drift with moon phase, temperature, food availability, or the animal habituating to (or growing wary of) the camera.

Groups who only recite the four assumptions without engaging with which ones are actually shaky haven’t fully answered 1a – push them toward independence/constant- p specifically, since that’s what 2e circles back to ask for a concrete example of.

$N = 8, p = 0.3, k = 3$; all figures below are pre-rendered in the handout (no code for students to run).

```
N <- 8; p <- 0.3; k <- 3
dbinom(k, N, p)
```

```
## [1] 0.2541218
```

```
which.max(dbinom(0:8, 8, 0.3)) - 1 # mode, 0-indexed
```

```
## [1] 2
```

Answer key: $P(X = 3) = 0.2541$; mode = 2.

Poisson comparison, part A ($N = 8, p = 0.3$ vs. $\lambda = 2.4$): binomial variance $Np(1 - p) = 1.68$; Poisson variance $\lambda = 2.4$ – mismatched, and the pre-rendered barplots look visibly different (Poisson has a fatter right tail).

Poisson comparison, part B ($N = 100, p = 0.024$ vs. $\lambda = 2.4$): binomial variance $= 100(0.024)(0.976) = 2.3424$ – close to 2.4, and the barplots should look nearly identical. Rule of thumb (2d): *Poisson is a good stand-in for binomial when N is large and p is small* (so $(1 - p) \approx 1$ and $Np(1 - p) \approx Np = \lambda$).

Watch for: groups who conclude Poisson is “just always fine” from Scenario B alone without noticing Scenario A disagreed – push them back to the variance mismatch in Scenario A as the reason it isn’t universal.

Act 3: continuous body mass ($\mu = 4.5, \sigma = 0.6$), via z-table

```
mu <- 4.5; sigma <- 0.6
z <- (5.5 - mu) / sigma
z

## [1] 1.666667
1 - pnorm(round(z, 2))           # table-precision version of 1 - Phi(z)

## [1] 0.04745968
1 - pnorm(5.5, mu, sigma)      # exact, for comparison

## [1] 0.04779035
mu + 1.28 * sigma              # table-lookup version of the 90th percentile

## [1] 5.268
qnorm(0.9, mu, sigma)          # exact, for comparison

## [1] 5.268931
```

Answer key: $z = (5.5 - 4.5)/0.6 = 1.67$ (table row 1.6, column .07); $\Phi(1.67) \approx 0.9525$, so $P(\text{mass} > 5.5) \approx 0.0475$ ($\approx 5\%$; the exact value via `pnorm` is 0.0478 – table rounding accounts for the small difference). For 3c, the table’s closest entry to 0.90 is $\Phi(1.28) = 0.8997$, giving a cutoff of $4.5 + 1.28(0.6) \approx 5.27$ kg (exact value via `qnorm` is 5.27 kg – matches).

Watch for: 1b/1c (density vs. probability) is the conceptual crux of Part 1 – a group that answers “ $P(\text{mass} = 4.5)$ is very small but not zero” hasn’t fully separated density from probability yet. Point them back at the shaded-area picture from 1a: a single point has zero *width*, hence zero area, hence zero probability, regardless of how tall the curve is there. In Part 3, watch for groups who forget to convert the z-score back to a body mass in 3c (the table gives z , not x).

PART 2 – CHEAT SHEET (circulating)

Copy for circulating during lab. Full reasoning and code: see Part 1 above.

Opening (~5 min)

- Two rules only: **addition** (disjoint outcomes sum) and **multiplication/rate-application** ($\text{rate} \times \text{count} = \text{expected count in a subgroup}$)
- Don't derive Bayes' rule or use $P(A | B)$ notation – Act 1 gets there on its own; Wednesday names it

Act 1 – classifier PPV (discovery)

- Part 1 (count, $N = 100\text{k}$): flagged/true ≈ 10 , flagged/false $\approx 1,000$, total flagged $\approx 1,010 \rightarrow \text{PPV} \approx 1\%$
- Part 2 hint if stuck: “what did you multiply by 100,000 in Part 1? Do that again with N instead”
- Part 2 answer: $\text{PPV} = \frac{\text{prev} \times \text{sens}}{\text{prev} \times \text{sens} + (1 - \text{prev}) \times \text{fpr}} \approx 0.0097$
- Part 3 (Site B, prevalence = 0.01): $\text{PPV} \approx 50\%$ – prevalence dominates
- **Watch for:** $\text{PPV} > 0.5$ at Site A usually means $P(S)/P(F)$ flipped

Act 2 – binomial vs. Poisson

- $N = 8, p = 0.3, k = 3$: hand calc = 0.2541; mode = 2
- Var mismatch ($N = 8, p = 0.3$): binom 1.68 vs. Pois 2.4 – **don't match**
- Var match ($N = 100, p = 0.024$): binom 2.3424 vs. Pois 2.4 – **do match**
- Rule of thumb: Poisson $\approx$ binomial when N large, p small

Act 3 – continuous mass, via z-table

- $z = (5.5 - 4.5)/0.6 = 1.67 \rightarrow \Phi(1.67) \approx 0.9525 \rightarrow P(\text{mass} > 5.5) \approx 0.048$
- Table's closest entry to 0.90 is $z \approx 1.28 \rightarrow \text{cutoff} \approx 4.5 + 1.28(0.6) = 5.27$ kg
- **Watch for:** “small but nonzero” answers to $P(X = 4.5)$ (push back to zero width $\rightarrow$ zero area); forgetting to convert z back to x in 3c